Harness



File: harness.kicad_sch

Daughterboard



File: daughter.kicad_sch

Wideband, MAP, USB, AUX



File: io.kicad_sch

Miata ECU motherboard 2001–2005 (NB2, VVT). Carries the year-specific harness header and the year-independent I/O (wideband, MAP, IAT, USB-C). All engine control is on the daughterboard.

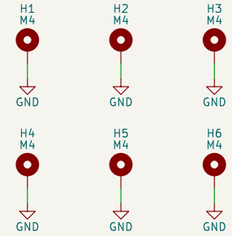
Sheet: /
File: miata_mb_0105.kicad_sch

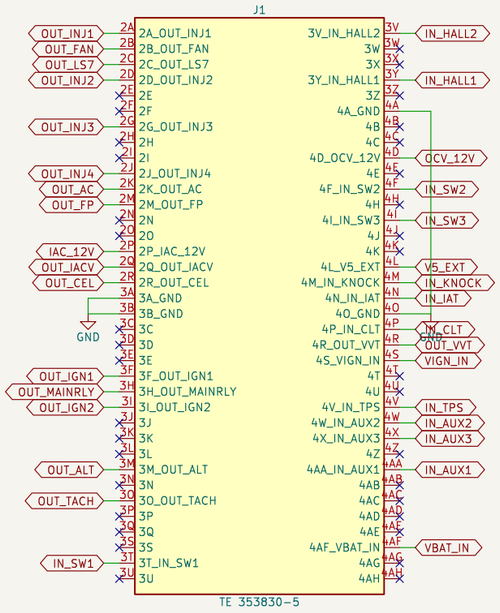
Title: Miata ECU motherboard 2001–2005 (NB2, VVT)

Size: A4	Date: 2026-10-30	**Rev: 0.9**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/4

Stock harness header (Miata ECU motherboard 2001–2005): every pin labelled with the daughterboard signal it drives

Case mounting bosses (M4), tied to ground





Generated from scripts/mb_nb2.py; netlist identical to the routed board

Sheet: /Harness/
File: harness.kicad_sch

Title: Miata ECU motherboard 2001–2005 — Harness

Size: A3	Date: 2026-10-30	**Rev: 0.9**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/4

Daughterboard sockets: pin-for-pin with the daughterboard J1 (top) and J2 (bottom)

Daughterboard standoffs (M3)

HD1 M3 standoff

HD2 M3 standoff

HD3 M3 standoff

HD4 M3 standoff



Generated from scripts/mb_nb2.py; netlist identical to the routed board

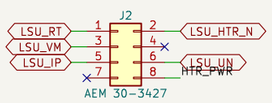
Sheet: /Daughterboard/
File: daughter.kicad_sch

Title: Miata ECU motherboard 2001–2005 — Daughterboard

Size: A3	Date: 2026-10-30	**Rev: 0.9**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/4

Wideband sensor harness AEM 30-3427 (controller on the daughterboard)

AEM pin: 1 trim (RT) 2 H− 3 common (VM) 5 pump (IP) 6 sense (UN) 8 H+. JST PUD S08B-PUDSS-1 (plug PUDP-08V-S)

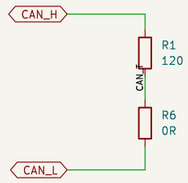




Heater fuse: 2 A hold, 30 V PTC

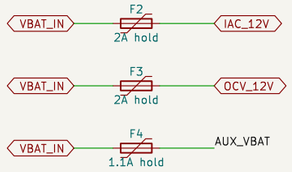
IP/VM/RT/UN and H− run to daughterboard J2 33–38; the heater current returns on J2 39/40

CAN bus termination at the ECU end

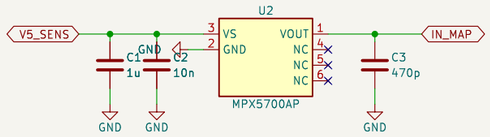


120R; remove R6 if the bus is terminated elsewhere

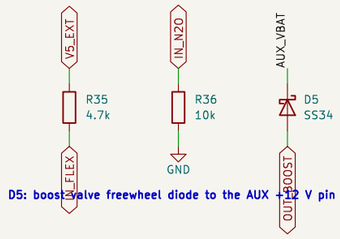
Fused +12 V feeds (harness actuators, AUX)



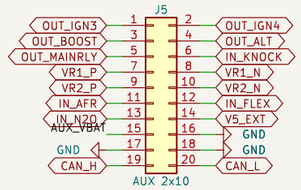
MAP sensor



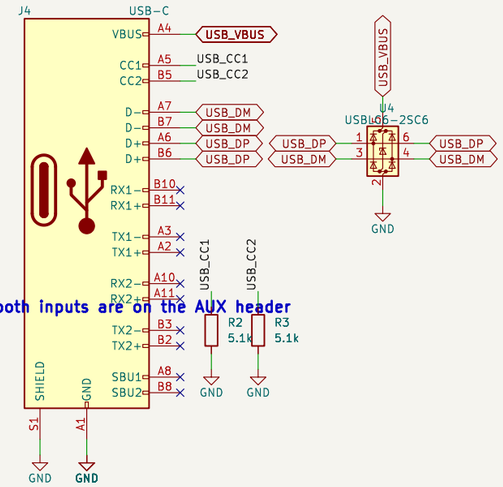
Flex-fuel sensor pull-up (5 V) and nitrous arming pull-down; both inputs are on the AUX header



D5: boost-valve freewheel diode to the AUX +12 V pin

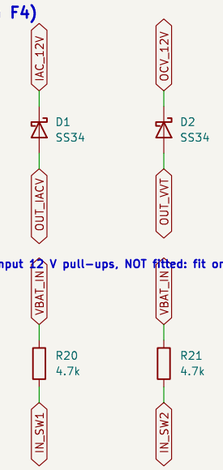


USB-C (device): tuning, logs, DFU firmware update

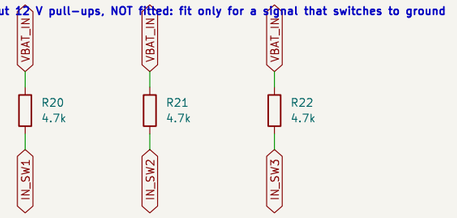


Valve freewheel diodes (to each valve's +12 V feed) and fit options

AUX: add-on I/O the stock harness has no pins for (+12 V through F4)



Switch-input 12 V pull-ups, NOT fitted: fit only for a signal that switches to ground



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Sheet: /Wideband, MAP, USB, AUX/
File: io.kicad_sch

Title: Miata ECU motherboard 2001–2005 — Wideband, MAP, USB, AUX

Size: A3	Date: 2026-10-30	**Rev: 0.9**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/4